

| Statistical Topic                                  | Category (if applicable)               | Assigned for Slide |
|----------------------------------------------------|----------------------------------------|--------------------|
| Randomized Controlled Trials (RCTs)                | Design                                 |                    |
| Factorial Designs                                  | Design                                 | Naaveen            |
| Propensity Score Matching                          | Design / Causal Inference              | Vy                 |
| Survey Sampling                                    | Design                                 |                    |
| Principal Component Analysis                       | Dimension Reduction                    | Vanesa             |
| Factor Analysis                                    | Multivariate Data Analysis             | Vy                 |
| Discriminant Analysis (Linear and Quadratic)       | Machine Learning / Supervised Learning | Ebony              |
| Boosting (further investigation)                   | Machine Learning                       | Naaveen            |
| Support Vector Machines                            | Machine Learning                       | Daizy              |
| k-Nearest Neighbors                                | Machine Learning                       | Daizy              |
| Nonparametric CIs for Proportions                  | Nonparametrics                         | Alex               |
| Wilcoxon Rank Sum + Kruskal-Wallis                 | Nonparametrics; Hypothesis Testing     | Donna              |
| Kernel Density Estimation                          | Nonparametrics                         | Isabel             |
| Rank-Based (or nonparametric) Regression           | Nonparametrics                         |                    |
| LOESS                                              | Nonparametrics                         | Donna              |
| ARIMA Models                                       | Time Series                            | Cody               |
| GARCH Models                                       | Time Series                            | Cody               |
| Spatial Analysis (General)                         |                                        | Rania              |
| Kriging                                            | Spatial Analysis                       | Rania              |
| Latent Dirichlet Allocation (LDA)                  | Text Analysis                          | Isabel             |
| Sentiment Analysis                                 | Text Analysis                          |                    |
| Cox Proportional Hazards Model                     | Survival Analysis                      | Joshua             |
| Quantile Regression                                | Regression                             |                    |
| Mixed Effect Models                                | Regression                             | Alex               |
| Bootstrap CIs                                      | Bootstrapping                          | Rashha             |
| Multidimensional Scaling + related (LLE)           | Dimension Reduction                    |                    |
| Canonical Correlation Analysis                     | Dimension Reduction                    |                    |
| Generalized Additive Models                        | Regression                             | Miranda            |
| Structural Equation Models                         | Multivariate Data Analysis             | Joshua             |
| AB Testing (if deep into it) or Sequential Testing | Hypothesis Testing                     | Miranda            |
| False Discovery Rates (further info)               | Hypothesis Testing                     |                    |
| Bayesian Modeling (rSTAN package)                  | Bayesian Methods                       | Vanesa             |
| SIR Model (Infectious Disease Spread)              | Epidemiology                           |                    |

|                                                |                                                    |         |
|------------------------------------------------|----------------------------------------------------|---------|
| Network Statistics / Analysis                  | Networks                                           | Rashha  |
| Network Sampling                               | Networks                                           |         |
| Network Plotting                               | Networks                                           |         |
| Longitudinal Data Analysis \ Repeated Measures |                                                    |         |
| Functional Data Analysis                       |                                                    |         |
| Changepoint Detection                          |                                                    |         |
| Cluster Analysis (many algorithms)             | Multivariate Data Analysis (Unsupervised Learning) | Jack    |
| Latent Class Analysis                          | Multivariate Data Analysis                         |         |
| Hierarchical Models                            | Regression                                         |         |
| Deep Learning (General)                        | Neural networks (Machine Learning)                 |         |
| Generative Adversarial Network                 | Neural networks (Machine Learning, AI)             | Jack    |
| Convolutional Neural Nets                      | Neural networks (Machine Learning)                 | Ebony   |
| LSTMs (Recurrent Neural Nets)                  | Neural networks (Machine Learning)                 | Tucker  |
| Causal Inference (General)                     |                                                    | Chelsea |
| Rare Event Modeling                            | Modeling                                           |         |
| Semiparametric Modeling                        | Machine Learning (Semi-supervised Learning)        |         |
| Permutation Methods (in new scenarios)         |                                                    |         |
| Applications of Markov Chains                  | Various                                            |         |
| Graphical LASSO                                | LASSO on networks                                  |         |
| Covariance Estimation                          | Estimation                                         |         |
| Hidden Markov Models                           | Modeling                                           | Chelsea |
| MICE (Multiple Imputation Chained Equations)   | Missing Data                                       |         |
| Missing Data                                   |                                                    | Tucker  |
| Pairwise Comparisons (Dependent)               | Nonparametrics (Sports Analysis in particular)     |         |